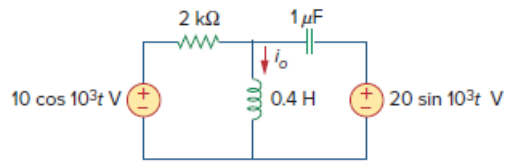


Problem

Use mesh analysis to find current i_o in the circuit of Fig. 10.74.

Figure 10.74:



Step-by-step solution

Step 1 of 6

Refer to Figure 10.74 in the text book.

The voltage sources values are,

$$v_1(t) = 10 \cos 10^3 t \text{ V}$$

$$\mathbf{V}_1 = 10 \angle 0^\circ \text{ V}$$

And,

$$v_2(t) = 20 \sin 10^3 t \text{ V}$$

$$= 20 \cos(10^3 t - 90^\circ) \text{ V}$$

$$\mathbf{V}_2 = 20 \angle -90^\circ \text{ V}$$

Calculate the impedance value of inductor.

$$\mathbf{Z}_L = j\omega L$$

$$= j(10^3)(0.4)$$

$$= j400 \Omega$$

Calculate the impedance value of capacitor.

$$\mathbf{Z}_C = \frac{1}{j\omega C}$$

$$= \frac{1}{j(10^3)(1 \times 10^{-6})}$$

$$= -j1000 \Omega$$

Step 2 of 6

Draw the phasor equivalent circuit as shown in Figure 1.

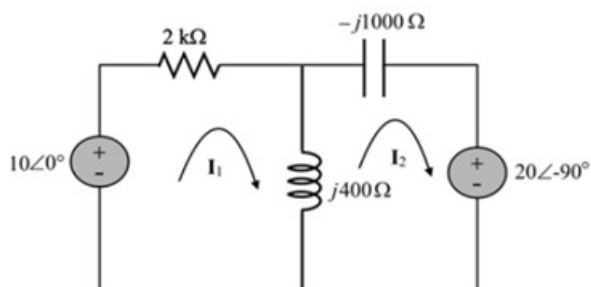


Figure 1

Step 3 of 6

Apply Kirchhoff's voltage law to loop 1.

$$-10 + 2000\mathbf{I}_1 + j400(\mathbf{I}_1 - \mathbf{I}_2) = 0$$

$$(2000 + j400)\mathbf{I}_1 - (j400)\mathbf{I}_2 = 10 \dots\dots (1)$$

Apply Kirchhoff's voltage law to loop 2.

$$j400(\mathbf{I}_2 - \mathbf{I}_1) - j1000\mathbf{I}_2 + 20\angle -90^\circ = 0$$

$$-j400\mathbf{I}_1 + (j400 - j1000)\mathbf{I}_2 + 20\angle -90^\circ = 0$$

$$j400\mathbf{I}_1 + j600\mathbf{I}_2 = 20\angle -90^\circ \dots\dots (2)$$

Express equations (1) and (2) in matrix form.

$$\begin{bmatrix} 2000 + j400 & -j400 \\ j400 & j600 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -j20 \end{bmatrix}$$

Step 4 of 6

Calculate Δ .

$$\begin{aligned} \Delta &= \begin{vmatrix} 2000 + j400 & -j400 \\ j400 & j600 \end{vmatrix} \\ &= (2000 + j400)(j600) - (-j400)(j400) \\ &= -240000 + j1200000 - 160000 \\ &= -400000 + j1200000 \end{aligned}$$

Calculate Δ_1 .

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 10 & -j400 \\ -j20 & j600 \end{vmatrix} \\ &= (10)(j600) - (-j20)(-j400) \\ &= 8000 + j6000 \end{aligned}$$

Determine the mesh current $\mathbf{I}_1$.

$$\begin{aligned} \mathbf{I}_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{8000 + j6000}{-400000 + j1200000} \\ &= 2.5 - j7.5 \text{ mA} \end{aligned}$$

Step 5 of 6

Calculate Δ_2 .

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 2000 + j400 & 10 \\ j400 & -j20 \end{vmatrix} \\ &= (2000 + j400)(-j20) - (10)(j400) \\ &= -j40000 + 8000 - j4000 \\ &= 8000 - j44000 \end{aligned}$$

Determine the mesh current $\mathbf{I}_2$.

$$\begin{aligned} \mathbf{I}_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{8000 - j44000}{-400000 + j1200000} \\ &= -35 + j5 \text{ mA} \end{aligned}$$

Determine the current $\mathbf{I}_o$.

$$\mathbf{I}_o = \mathbf{I}_1 - \mathbf{I}_2$$

Substitute $2.5 - j7.5$ mA for $\mathbf{I}_1$ and $-35 + j5$ mA for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_o &= 2.5 \times 10^{-3} - j7.5 \times 10^{-3} - [-35 \times 10^{-3} + j5 \times 10^{-3}] \\ &= 2.5 \times 10^{-3} - j7.5 \times 10^{-3} + 35 \times 10^{-3} - j5 \times 10^{-3} \\ &= 37.5 - j12.5 \text{ mA} \\ &= 39.53 \angle -18.43^\circ \text{ mA}\end{aligned}$$

The expression of $i_o(t)$ is,

$$i_o = 39.53 \cos(10^3 t - 18.43^\circ) \text{ mA}$$

Therefore, the expression of i_o is $\boxed{39.53 \cos(10^3 t - 18.43^\circ) \text{ mA}}$.